



CIBOCOMPASS

Navigating Italian Cuisine for International
Students

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PROJECT OUTLINE

This report details the development of "CiboCompass," a mobile application designed to assist international students in Italy with understanding local cuisine from a nutritional and cultural perspective. The project addresses the unique challenges faced by international students in navigating unfamiliar restaurant menus, particularly in terms of dish ingredients, cultural taste preferences, and evaluating meals based on community-based ratings.

NEED FINDING

Need finding Phase

The idea for *CiboCompass* was deeply rooted in our own lived experiences. As a team of six international students studying in Italy, we had each personally encountered challenges when trying to navigate local restaurant menus. Unfamiliar dish names, unknown ingredients, and uncertainty about whether a dish would align with our dietary habits or taste preferences were common sources of confusion. These personal frustrations sparked our initial hypothesis: that such difficulties were not unique to us, but shared by the broader international student community.

To validate this assumption, we began our need finding phase with **20 structured interviews**^[1] each conducted with international students from diverse cultural and dietary backgrounds. These interviews, recorded with full consent, uncovered strong recurring themes:

- A reliance on **Google Translate/Lens** to decipher menus
- Frustration with the lack of **dish pictures or ingredient breakdowns**
- Frequent **dietary concerns** linked to religion, culture, or health (e.g., halal, vegetarian, allergens)
- A tendency to trust food suggestions from peers of **similar cultural backgrounds**

Insights from these interviews directly informed the design of our survey^[2], which was then distributed and completed by over **170 participants**. After cleaning the data, we retained **140 valid responses**. The survey quantitatively reinforced what the interviews suggested:

- Over **40%** of respondents rated **ingredient lists** as the most useful

app feature

- Roughly **70%** said they would value **peer-based reviews from users with the same background**
- A significant number expressed the need for **dietary filters** such as halal, vegan, or allergen alerts
- **Calorie and fitness features** were appreciated by many, but flagged as potentially anxiety-inducing—indicating the ice of **optional visibility**

Competitor Analysis

In parallel, we conducted a **competitor analysis**^[3]—first examining well-known nutrition apps like *MyFitnessPal* and *Lifesum*, and later expanding our view based on user interviews. Unexpectedly, many students relied more on **Google Maps, Instagram, Xiaohongshu, and HappyCow** for discovering food and judging its compatibility with their preferences. However, these tools lacked an integrated approach that combined **translation, cultural sensitivity, ingredient-level transparency, and feedback personalization**.

This convergence of **lived experience, qualitative interviews, quantitative survey data, and market research** laid the foundation for our first prototype—one designed not just to inform, but to empower international students in Italy with culturally meaningful, context-aware food choices.

From Initial Concept to User-Centered Focus

The development of *CiboCompass* began with a feature-driven approach shaped by the course theme of *fitness and nutrition*. Our initial concept focused on three main functions:

1. **Calorie Estimation and Fitness Suggestion** – Users could input the name of an Italian dish, and the app would estimate its calorie content and suggest the amount of running time required to burn it, highlighting the fitness dimension.
2. **Dish Composition Display** – The app would present a breakdown of the dish's ingredients and components to support better understanding and dietary awareness.

3. **User Taste Analysis by Nationality** – Based on aggregated user feedback, the app would display how much people from the user’s cultural background typically liked the dish.

While this version of the concept provided a broad functional foundation, our first review with the professor challenged us to narrow the scope by identifying three distinct and meaningful user tasks. This led us to redefine the app’s focus through a more experience-oriented lens, translating it into three essential user-centered questions:

- **“What’s this dish? Can I eat it?”**
- **“What do I think about it afterward?”**
- **“Do people like this in my culture?”**

These tasks shifted our attention from feature listing to user goals, laying the foundation for the need-finding phase that followed.

^[1] ^[2] ^[3] **NOTE:** For a comprehensive account of the methods, data, and insights gathered during this phase, we encourage readers to consult the full-length supporting documents provided in the project folder, called:

□Interview_Summary_Report

□Interview_Supports_Survey_Design

□CiboCompass_Report_Updated

□Competitor_Analysis_Report

Interview_questions_adapted

Storyboarding

To better communicate the user journey and emotional motivations behind *CiboCompass*, the team developed a series of illustrated **storyboards**^[4].

The storyboard development was informed by insights collected from user interviews. Common user needs that surfaced included:

- Concern over dietary restrictions (e.g., halal, vegetarian, allergies) and difficulty of understanding dish details

- Desire for reassurance through culturally relevant feedback
- Desire of contributing feedback for an eaten dish

Four core scenarios were developed, three of them kept and one aborted after the first revision, each mapped into a **4-panel comic strip** using a hybrid method: AI-generated drafts (via ChatGPT image tools) were refined in **Procreate** to add facial expressions, clear UI depictions, and consistent design language.

STORYBOARD 1 *WHAT'S THIS DISH, CAN I EAT IT?*



A student sits in front of a menu with no pictures. He uses the app to search the name of a dish, sees a photo and description, and check whether it contains restricted ingredients (e.g., pork). If unsafe, the app suggests an alternative.

Figure 1 Storyboard 1 final
Storyboard 1 Evolution



Figure 2 Storyboard 1 version 1



Figure 3 Storyboard 1 version 2

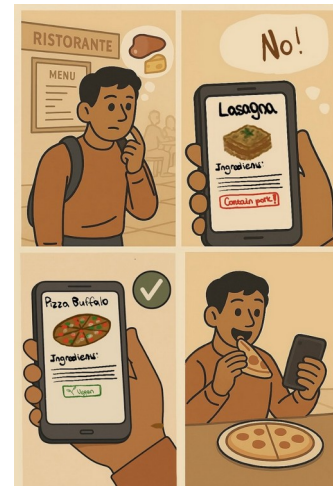


Figure 4 Storyboard 1 final

STORYBOARD 2 *DO PEOPLE LIKE THIS IN MY CULTURE?*



Figure 5 Storyboard 2 final
Storyboard 2 Evolution

A user checks a dish and sees that **people from their country recommended this dish**, which helps them decide to try it.

This provides social reassurance and leverages cultural familiarity as a decision-making tool.



Figure 6 Storyboard 2 version 1



Figure 7 Storyboard 2 version 2



Figure 8 Storyboard 2 final

STORYBOARD 3 *WHAT DO I THINK ABOUT IT AFTERWARD?*



After finishing a dish (e.g., *Pizza Buffalo*), the student rates it within the app, contributing to the cultural feedback loop.

This panel highlights community-building, where users move from passive readers to active contributors

Figure 9 Storyboard 3 final
Storyboard 3 Evolution



Figure 10 Storyboard 3 version 1



Figure 11 Storyboard 3 version 2

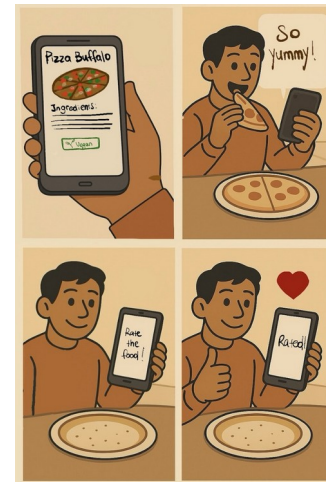


Figure 12 Storyboard 3 final

For Each **Storyboard**, several modifications were made. First, due to inaccuracies in AI-generated content—such as incorrect spelling or textual errors—we manually corrected the text. Compared to the first version, the second iteration adopted a more visual approach by incorporating additional graphical symbols and reducing reliance on textual descriptions.

The evolution from **version 2 to final** was based on feedback received during the second revision with the professor. Specifically, it was suggested that the ingredient text was too small to read and not essential at this stage. As a result, we replaced the detailed text with **placeholder sketch lines**, which allowed us to shift visual emphasis toward more storytelling elements.

Visual & Interaction Design Notes

- Each panel focused on a distinct emotional or action moment
- Visual symbols like , ,  were used to enhance quick understanding
- A consistent tone and character style was maintained across all storyboards

^[4] **NOTE:** All storyboard panels and related analysis reports can be found in the “Storyboard” folder under NeedFinding directory.

PROTOTYPING CYCLES

Paper prototyping

The team initiated the prototyping phase by developing and testing low-fidelity paper prototypes^[5]. This early-stage design approach allowed us to rapidly explore user interface ideas and interaction flows with minimal resource investment.

To evaluate usability and gather first impressions, we conducted **user testing sessions with 5 participants**. Each participant was given a contextual backstory to immerse them in the experience:

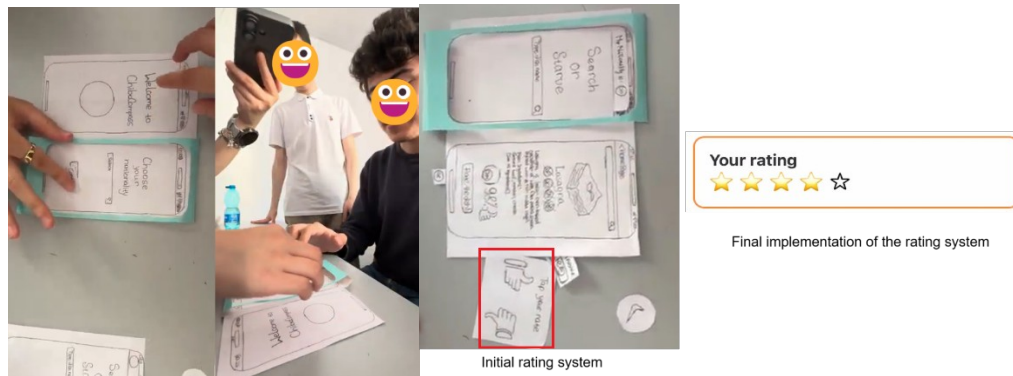
"You found this app called CiboCompass on the AppStore, and the app description claimed that it helps international students identify dishes and their ingredients to guide them through Italian cuisine. After downloading the app, you have just opened it and are now on the home page."

The entire app flow was walked through using hand-drawn screen sketches. We recorded two camera angles during the tests (with participant consent explicitly recorded on video before starting), allowing us to capture both facial reactions and hand interactions. As participants navigated through the interface, we observed points of confusion, hesitation, or smooth interaction, and took detailed notes accordingly.

The testing surfaced several usability insights:

- Certain features were intuitively understood and required no redesign—these were marked for visual refinement only.
- Other areas caused noticeable confusion, particularly regarding the feedback mechanism.
- A key suggestion repeatedly mentioned was the **replacement of the binary thumbs-up/thumbs-down system with a more nuanced five-star rating scale**, which we later implemented in the digital prototype.

These tests validated several core flows while highlighting subtle pain points, confirming the importance of early feedback before digital commitment. This round of paper prototyping set a strong foundation for the subsequent digital design iterations.



^[5] **NOTE:** For a complete view of the paper prototyping process, including annotated user flows and participant interaction, please refer to the “*PaperPrototyping*” subfolder within the *Prototyping* directory. The folder contains model photos and testing materials used during this phase.

Figma Prototyping

Building on the insights gathered from paper prototyping and user testing, the team transitioned to digital prototyping using **Figma**. The first version of the interface incorporated concrete feedback from earlier sessions—particularly the introduction of a **five-star rating system**, clarified navigation flows, and enhanced visual hierarchy to better support user orientation.

The **first Figma prototype**^[6] was subjected to an **Expert Cognitive Walkthrough (CW)**, conducted with the assistance of **CW GPT**. The expert review surfaced several **minor usability issues**, along with targeted recommendations:

- **Low contrast in buttons and labels:** Improved for WCAG accessibility compliance
- **Premature flag update before "Save" is tapped:** Addressed by deferring updates until confirmation or adding a visual cue
- **“Unrate” button lacks confirmation:** Enhanced with a confirmation prompt or undo option
- **“Compare by nationality” feature:** Rated as clear, discoverable, and effective (no issue identified)

All issues identified were promptly resolved in the following version of the prototype.

Version 1, was presented during the **second review session with the**

course professor, where we also shared our paper prototyping results and user testing feedback.

During the review, the professor provided a series of highly actionable suggestions to enhance usability and immediacy:

- **Remove the initial search-only screen** and instead present a **default dish view directly** to let users interact with core features immediately, minimizing unnecessary friction.
- **Make the rating system more visible** by displaying the user's given rating explicitly, ensuring clarity about how user feedback is captured and represented.
- **Use a tabular format for ingredient display**, improving scannability and structured readability.
- **Change allergen indicators**, which were previously shown using emojis, were converted into **text-based tags** to avoid ambiguity and ensure clear communication.

These suggestions were fully adopted in subsequent iterations. In **Figma Version 2**, the homepage was streamlined, removing the standalone search screen and integrating interactive content from the start. Rating feedback was made more explicit, and ingredients were restructured using a clean, tabular layout in **Figma Version 3**, which passed the cognitive walkthrough evaluation without any further usability concern.

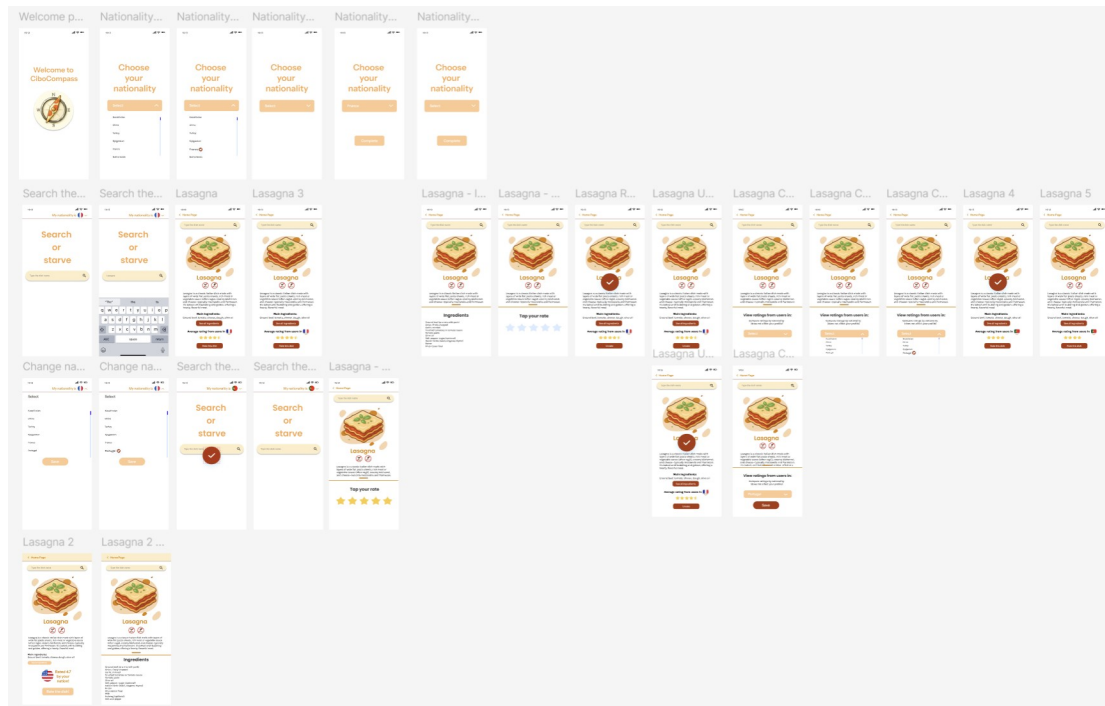


Figure 13 Figma Prototyping version 1

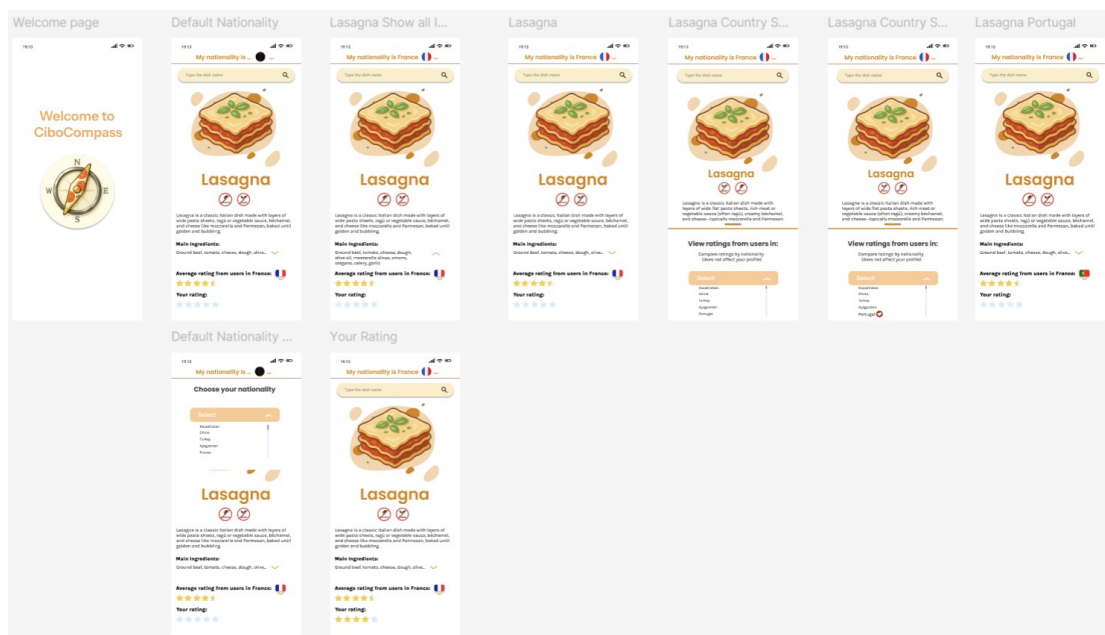


Figure 14 Figma Prototyping version 2

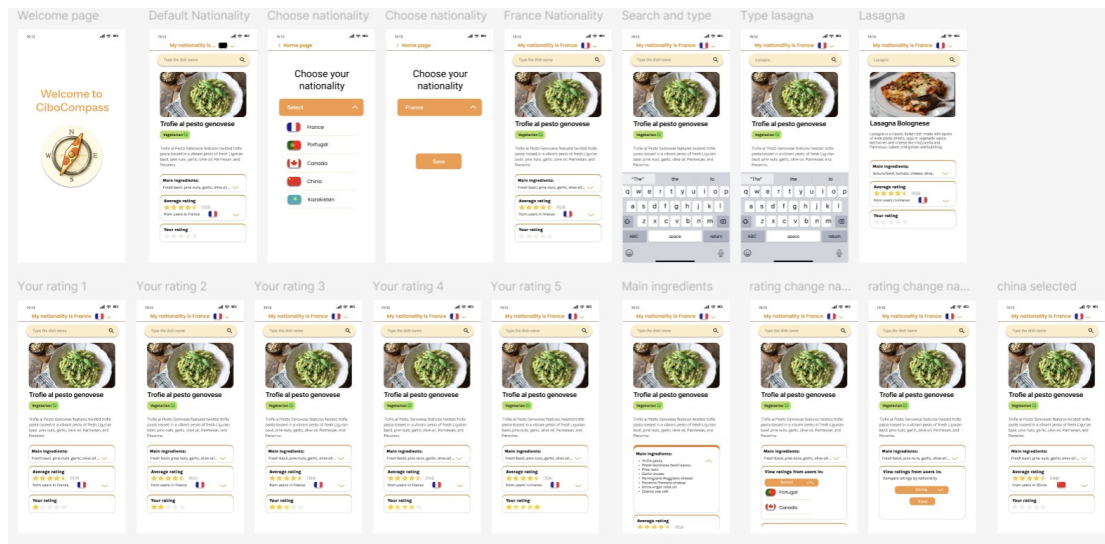


Figure 15 Figma Prototyping version 3

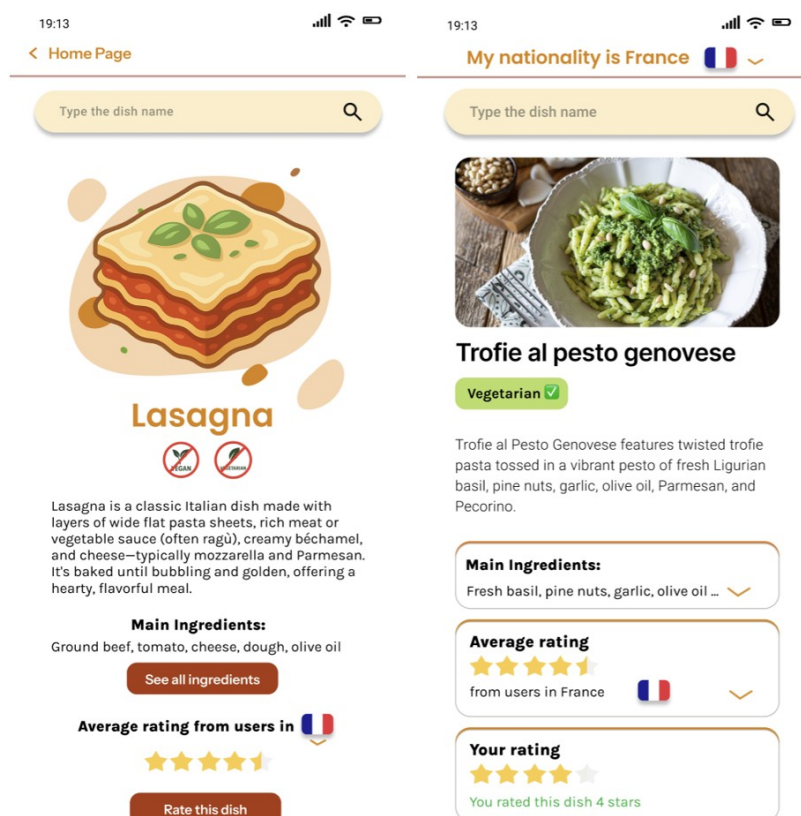


Figure 16 Figma prototyping version 1 - version 3 main feature comparison

No Figma Prototype was made for the new edition of the app.

[6] **NOTE:** Detailed CW reports are available in the “Prototyping” folder for further reference, the figma protope project is accessible through: [this link](#).

Responsive application front end

Following the design refinements made in Figma, the final interactive version of *CiboCompass* was implemented using **React Native**^[6], ensuring a fully responsive interface across both iOS and Android devices. It's the final prototype that will be shown during the exam.

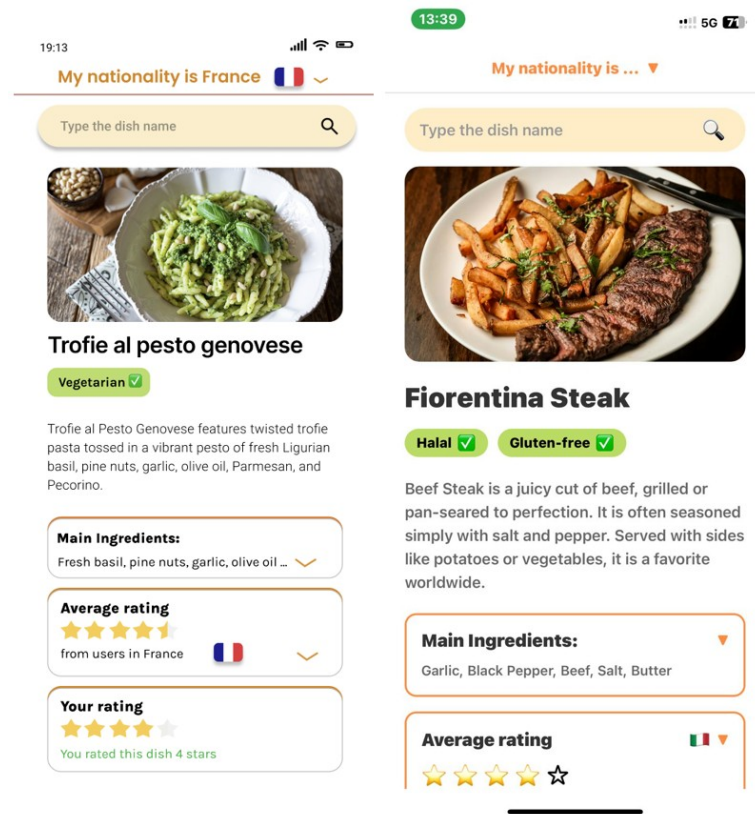


Figure 17 Figma design vs React Native main feature comparison

We made a conscious effort to **faithfully replicate the Figma design** in the React Native implementation. While most of the UI and interactions matched closely, **minor visual and behavioral differences** were inevitable due to framework constraints and rendering variations across platforms.

A small-scale **usability test** was conducted to validate the performance and accessibility of this responsive version. Participants interacted directly with the deployed React Native app, and feedback was collected to assess both user satisfaction and interface performance.

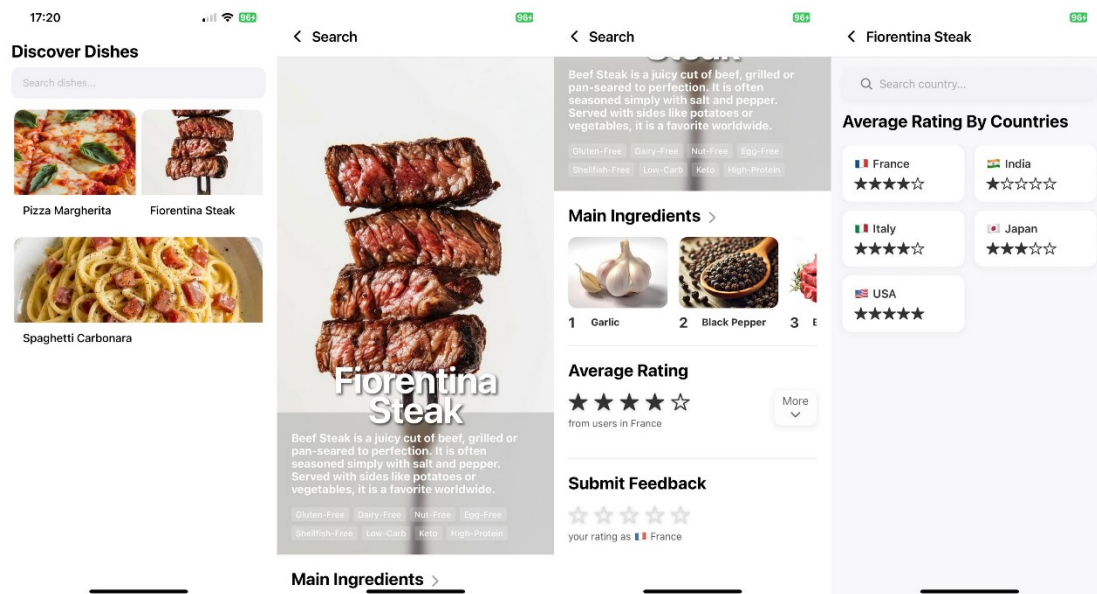


Figure 18: New iOS-centric React Native app design showcasing key features.

The development of our second version (after the first exam) focused on achieving a distinct iOS aesthetic and interaction paradigm, departing significantly from previous iterations. This involved leveraging native iOS UI components such as large, clear titles in navigation bars, segmented controls, and prominent action buttons. This approach prioritizes a clean, minimalist aesthetic with ample whitespace. We took inspiration from the Apple TV app. While aiming for a near-native iOS experience, we conscientiously maintained a single codebase using React Native, which inherently provides a degree of cross-platform compatibility, though our primary design focus and optimization efforts were directed towards achieving an authentic iOS feel.

A small-scale usability test was conducted with **seven participants (five valid participants)** to validate the performance and user experience of this updated version. Participants interacted directly with the deployed React Native app on iOS devices, and feedback was gathered to assess both user satisfaction and interface performance.

Key Findings:

- Users generally found the app's intuitive layout and clean interface to be positive, noting smooth interactions and responsiveness.

- The dynamic display of culturally filtered ratings was well-received and considered helpful for guiding food decisions.
- The tag-based classification system, including Vegetarian, Vegan, Halal, Kosher, and Gluten-Free, was particularly useful for users with dietary restrictions. Stephen from the Philippines specifically liked the allergen indicators.
- The large dish image served as a strong focal point, and the horizontal list was easily understood by users like Stephen from the Philippines.
- Users found the search functions intuitive.

Some Unique Feedback for Future Updates:

- **"Clicked on countries in the country average rating page"**
functionality: Initially, this action did not yield a result. After user testing, this has been fixed, and now provides appropriate functionality when users click on countries in the average rating page.
- **Average ratings explanation:** Vanessa from China suggested that the average ratings could be explained more thoroughly and recommended using more meaningful explanatory labels.
- **Categories and Drinks Menu:** Amir from Iran suggested incorporating categories of dishes (e.g., pastas, pizzas) on the main screen and also proposed the addition of a drinks menu.
- **Comments feature:** G from Kazakhstan suggested adding a comment section for dish reviews. While this was a valuable suggestion, it was intentionally excluded from the current development scope to maintain feature focus.
- **Search filtering:** An anonymous user from South Korea suggested adding filtering options to the search bar on the main screen.

The app passed its usability test without functional issues or crashes, and core functionalities performed as intended.

^[6] **NOTE:** The software source code can be found in GitHub in the following link: <http://github.com/Diastart/cibo>

TECHNICAL DEVELOPMENT DETAILS

The technical implementation of *CiboCompass* was carried out in parallel across frontend and backend teams, with a focus on efficiency, modularity, and cultural adaptability.

Backend Development

The backend was implemented in **Go**, with three active API routes:

- GET /v1/dishes/{name} - fetches dish details from the database
- POST /v1/dishes/{name}/feedback - handles user feedback submission
- GET /v1/healthcheck - monitors server status

The server runs on port 4000 by default, uses httprouter for routing, and mattn/go-sqlite3 as the database driver. The database schema includes four normalized tables:

- **Dishes** - stores core dish data including images and descriptions
- **Ingredients** - lists standardized ingredients
- **DishesToIngredients** - maps many-to-many dish-ingredient relationships
- **Feedbacks** - stores user-submitted ratings filtered by nationality

Sample data was pre-loaded for popular Italian dishes such as *Pizza Margherita*, *Spaghetti Carbonara*, and *Fiorentina Steak*, with culturally differentiated feedback from users of nationalities like Italy, USA, Japan, India, and France. The API personalizes responses by using a **nationality header**, ensuring culturally relevant recommendations.

Frontend Development

On the frontend, we developed a **React Native app** using **Expo SDK 53** and **React 19**, supporting both iOS and Android. The app uses AsyncStorage for local data persistence and interacts with the Go backend over HTTP.

Implemented features include:

- **Nationality setup modal** at first launch
- **Real-time search** for dish names
- **Expandable ingredient lists**, now presented in **tabular format** for

clarity

- **Star-based rating system** with interactive visual feedback
- **Cultural rating displays**, showing how users from similar backgrounds rated each dish
- **Dietary specification badges**, dynamically identifying whether dishes are vegetarian, vegan, halal, kosher, or gluten-free through ingredient analysis
- **Smooth animations, error handling, and scroll state persistence** for a polished user experience

The app communicates with the backend via **JSON over HTTP**, sending the user's nationality in request headers. User preferences and rating history are locally stored using **AsyncStorage**, allowing for personalized and persistent behavior across sessions.

FINAL THOUGHTS

While the current version of *CiboCompass* successfully delivers its core functionalities, we recognize that the user interface still holds **significant room for improvement**. Potential future enhancements include:

- Adding a "**Recent Searches**" section to streamline repeated use
- Displaying the **number of ratings** a dish has received to improve transparency
- Introducing a **comment section** to allow users to share qualitative feedback
- Other user suggestions

Due to time constraints and the absence of a persistent **user session system** on the backend, some of these features were deferred. Their inclusion would require user tracking and storage mechanisms beyond the current implementation scope.

Throughout every phase of the project, **AI tools were consistently and effectively integrated** to accelerate development and enhance quality. Examples include:

- **Report writing:** merging and refining collaborative documents
- **Persona generation:** simulating realistic user types for design grounding
- **Survey optimization:** reviewing and improving questionnaire clarity
- **Storyboard creation:** drafting and illustrating user scenarios efficiently
- **Code debugging:** assisting in resolving frontend/backend implementation issues

Each instance of AI use has been **transparently documented** in the corresponding detailed reports, reflecting our commitment to both innovation and accountability.

This project not only allowed us to apply HCI principles in a culturally relevant context but also taught us valuable lessons in cross-functional teamwork, iterative design, and realistic development trade-offs under time pressure.